

SEBASTIÁN JOSÉ CARDONA RAMÍREZ

BACKEND ENGINEER • AI & DATA AUTOMATION

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PROFESSIONAL SUMMARY

Backend Engineer building async APIs and real-time data pipelines — actively moving into Data Engineering & AI Automation.

I design systems that ingest, process and serve data reliably at the API layer: FastAPI services handling live telemetry streams, PostgreSQL-backed pipelines, and event-driven architectures secured end-to-end (JWT, RBAC, Argon2id). My focus: turning real-world data — IoT sensors, business workflows — into structured, queryable, production-grade systems: the same discipline Data Engineering demands, applied from the backend layer.

TECHNICAL STACK

Backend / API: Python · FastAPI · async/await · REST · WebSockets

Data & Messaging: PostgreSQL · SQLAlchemy · Alembic · Redis · MQTT (event-driven ingestion)

Auth & Security: JWT · OAuth · RBAC · Argon2id

Infra & Deploy: Docker · Docker Compose · GitHub Actions (CI/CD) · AWS · Render

AI / Automation: Multi-agent orchestration · SDD (spec-driven development) · SRS-first workflows

Core Concepts: Event-driven architecture · ETL/ELT fundamentals · Idempotency · Data validation & schema design · CI/CD · 12-Factor App · CAP (distributed systems trade-offs)

EXPERIENCE

Mobile Development Research Assistant — UAN (Universidad Antonio Nariño)

Aug 2026 – Present

- Teach the Mobile Application Development course (App Inventor and React Native), using live coding as the primary teaching method.
- Mentor students in building real-world apps: components, navigation, state management, and API consumption.
- Cover data consumption, declarative and event-driven programming, and Apple HMI design principles.

FEATURED PROJECTS

VEYRA — Vehicle Inspection PWA • Specialized Vehicle Auditing Firm

May 2026 – Jul 2026

github.com/sebastiansaintt/car_checking

- Replaced paper-based fleet inspections with a full digital workflow: inspection history, maintenance tracking, and real-time fleet availability.
- Modular monolith (FastAPI + MVC) chosen deliberately over microservices to ship fast without premature complexity.
- Security-first: RBAC, one-way password hashing, principle of least privilege, HTTPS-only, server-side validation on every input.

Real-Time Fermentation Monitoring Pipeline • Craft Breweries

Nov 2025 – Apr 2026

github.com/sebastiansaintt/risk_follower

- Ingests live sensor data (temperature, humidity, CO₂) from ESP32 devices via MQTT (HiveMQ Cloud) into PostgreSQL for historical analysis.
- Authenticated REST API with role-based filtering (hardware, risk level, date); live WebSocket dashboard; JWT auth with 3 role tiers.
- Configurable range validation and audit logging on sensitive actions; deployed as a monolith + background worker on Render, with managed Postgres on Supabase.

LawSim SMB — Regulatory Simulation for Colombian SMEs

2025

- Hybrid architecture (Layers + Microservices + MVC); modeled Colombian regulations using OWL 2 ontologies (Protégé, Hermit), NLP (LEGALBETO), and SWRL rules for risk inference.

EDUCATION

Software Engineering — Universidad Cooperativa de Colombia (UCC), 2026

ADDITIONAL TRAINING

- Climate Change: From Learning to Action — United Nations, 2025
- Soft Skills Strengthening — HCL, HabComLearn, 2024
- Emotional Intelligence — CUN, 2022

LANGUAGES

Spanish (native) · English (advanced technical reading, basic-to-intermediate spoken communication)